Lecture 10

SU(2) and $\mathfrak{su}(2)$: highest-weight theory and spin

Objectives. *Lecture 9 ended with a licence: over* SU(2)*, group representations and algebra representations are the same thing. This lecture spends it. We set up the group and its algebra in the two standard bases and legalise complexification; we prove the classification theorem — exactly one irreducible representation of each dimension* $2j+1$*, with* J_z*-spectrum* $\{-j,\dots,j\}$ *and ladder coefficients derived rather than decreed; we realise every spin* j *concretely on homogeneous polynomials, complete with an explicit invariant inner product; we show that the Casimir operator equals* $j(j+1)$*; and we read the result back into physics, where it is the entire nonrelativistic theory of angular momentum and spin.*

10.1 The group, the algebra, and the two dictionaries

The calculation this lecture performs — complexify, find a highest weight, descend by ladders, count — is the model for every representation theory Part II will do. We first lay out the cast.

Proposition 10.1 (SU(2) is the three-sphere, officially)**.**

$$\mathrm{SU}(2) = \left\{ \begin{pmatrix} \alpha & \beta \\ -\bar\beta & \bar\alpha \end{pmatrix} :\ \alpha,\beta\in\mathbb{C},\ |\alpha|^2+|\beta|^2=1 \right\}, \tag{10.1}$$

so the map $U\mapsto(\alpha,\beta)$ *identifies* SU(2) *with the unit sphere* $S^3\subset\mathbb{C}^2\cong\mathbb{R}^4$.

Proof. For $U=\left(\begin{smallmatrix} a & b \\ c & d\end{smallmatrix}\right)$ with $\det U=1$, Cramer gives $U^{-1}=\left(\begin{smallmatrix} d & -b \\ -c & a\end{smallmatrix}\right)$, while unitarity demands $U^{-1}=U^\dagger=\left(\begin{smallmatrix} \bar a & \bar c \\ \bar b & \bar d\end{smallmatrix}\right)$. Equating: $d=\bar a$ and $c=-\bar b$; then $1=\det U=|a|^2+|b|^2$. Conversely any such matrix is unitary with determinant one. ∎

Lectures 8 and 9 used this identification three times — compactness, connectedness, simple connectivity — on the strength of Lecture 8's adjugate computation; the chart above makes it explicit.

Remark 10.2 (Quaternions)**.** The real span of $\mathbb{1}$, $-\mathrm{i}\sigma_1$, $-\mathrm{i}\sigma_2$, $-\mathrm{i}\sigma_3$ is closed under matrix multiplication, and the three imaginary units multiply like Hamilton's i,j,k — for instance $(-\mathrm{i}\sigma_1)(-\mathrm{i}\sigma_2)=-\mathrm{i}\sigma_3$. Writing $U=x_0\mathbb{1}-\mathrm{i}x_1\sigma_1-\mathrm{i}x_2\sigma_2-\mathrm{i}x_3\sigma_3$ reproduces (10.1) with $\sum x_\mu^2=1$: the group SU(2) is the unit quaternions, multiplying as matrices.

The Lie algebra is Lecture 8's: $\mathfrak{su}(2)$, with basis $X_a=-\frac{\mathrm{i}}{2}\sigma_a$ and brackets $[X_a,X_b]=\epsilon_{abc}X_c$; the physicist's Hermitian generators are $J_a=\mathrm{i}X_a=\frac{1}{2}\sigma_a$ with $[J_a,J_b]=\mathrm{i}\,\epsilon_{abc}J_c$ — the angular momentum commutation relations, read off from the dictionary of Lecture 8. Classification, however, happens in the complexification $\mathfrak{su}(2)_{\mathbb{C}} = \mathfrak{sl}(2,\mathbb{C})$ (computed in Lecture 8), and there a different basis does the work:

$$H = \begin{pmatrix} 1 & 0 \\ 0 & -1 \end{pmatrix}, \qquad E = \begin{pmatrix} 0 & 1 \\ 0 & 0 \end{pmatrix}, \qquad F = \begin{pmatrix} 0 & 0 \\ 1 & 0 \end{pmatrix},$$

with brackets $[H,E] = 2E$, $[H,F] = -2F$, $[E,F] = H$. The physics dictionary for this basis sets

$$J_z = \tfrac{1}{2}H, \qquad J_+ = E = J_x + \mathrm{i}J_y, \qquad J_- = F = J_x - \mathrm{i}J_y,$$

(the identities are immediate from the Pauli matrices), and the brackets become

$$[J_z, J_\pm] = \pm J_\pm, \qquad [J_+, J_-] = 2J_z. \tag{10.2}$$

The first relation says $J_\pm$ will shift J_z-eigenvalues by ± 1: they are *ladder operators*. Note what they are not: neither E nor F is anti-Hermitian, so the ladders do not belong to $\mathfrak{su}(2)$ at all. They live only in the complexification — and yet they act on every real-world spin system, by the following licence.

Lemma 10.3 (Complexification is legal)**.** *Let V be a finite-dimensional complex vector space. Every representation π of the real Lie algebra $\mathfrak{su}(2)$ on V extends uniquely to a complex-linear representation $\pi_{\mathbb{C}}$ of $\mathfrak{sl}(2,\mathbb{C}) = \mathfrak{su}(2) \oplus \mathrm{i}\,\mathfrak{su}(2)$, namely*

$$\pi_{\mathbb{C}}(X + \mathrm{i}Y) = \pi(X) + \mathrm{i}\,\pi(Y), \qquad X, Y \in \mathfrak{su}(2),$$

and every complex-linear representation of $\mathfrak{sl}(2,\mathbb{C})$ arises this way. A subspace is π-invariant if and only if it is $\pi_{\mathbb{C}}$-invariant; an operator intertwines two representations of $\mathfrak{su}(2)$ if and only if it intertwines their extensions. In particular π is irreducible if and only if $\pi_{\mathbb{C}}$ is.

Proof. The formula is well defined because $\mathfrak{su}(2) \cap \mathrm{i}\,\mathfrak{su}(2) = \{0\}$ (Lecture 8), and it is the only complex-linear extension. It preserves brackets: expanding $[X + \mathrm{i}Y, X' + \mathrm{i}Y'] = ([X,X'] - [Y,Y']) + \mathrm{i}\,([X,Y'] + [Y,X'])$ and comparing with the same expansion of $[\pi(X) + \mathrm{i}\pi(Y), \pi(X') + \mathrm{i}\pi(Y')]$ — legitimate because the commutator of operators is $\mathbb{C}$-bilinear — the two sides agree term by term since π preserves brackets. Conversely, restricting a complex-linear representation of $\mathfrak{sl}(2,\mathbb{C})$ to the real subalgebra $\mathfrak{su}(2)$ inverts the construction. The images $\pi(\mathfrak{su}(2))$ and $\pi_{\mathbb{C}}(\mathfrak{sl}(2,\mathbb{C}))$ have the same complex span, which proves the statements about invariant subspaces and intertwiners. ∎

This is why complexifying costs nothing and buys everything: the representations are the same, but $\mathfrak{sl}(2,\mathbb{C})$ owns the ladder operators, and $\mathbb{C}$ — being algebraically closed — owns eigenvectors. The classification can begin.

10.2 The classification: highest weights and ladders

Here is the theorem this lecture exists to prove. We adopt the physicist's habit of writing J_z, $J_\pm$ both for the elements of $\mathfrak{sl}(2,\mathbb{C})$ and for the operators representing them; the context disambiguates.

Theorem 10.4 (Highest-weight classification)**.** *Let π be an irreducible complex-linear representation of $\mathfrak{sl}(2,\mathbb{C})$ on V. Then there is a $j \in \{0, \frac{1}{2}, 1, \frac{3}{2}, \dots\}$ with $\dim V = 2j + 1$ such that J_z*

is diagonalisable with simple spectrum $\{j, j-1, \dots, -j\}$*, and* V *has a basis* $v_0, v_1, \dots, v_{2j}$ *on which*

$$J_z\, v_k = (j-k)\, v_k, \qquad J_-\, v_k = v_{k+1}, \qquad J_+\, v_k = k\,(2j-k+1)\, v_{k-1} \tag{10.3}$$

(with $v_{-1} = v_{2j+1} = 0$*). Two irreducible representations are equivalent if and only if they have the same* j *— equivalently, the same dimension. Conversely, for every* $j \in \frac{1}{2}\mathbb{Z}_{\geq 0}$ *such a representation exists; we call it the* spin-j representation π_j.

The theorem is one picture (Figure 10.1): a single column of J_z-eigenstates, the operators $J_\pm$ stepping between adjacent rungs and annihilating the two ends. The proof reads it off.

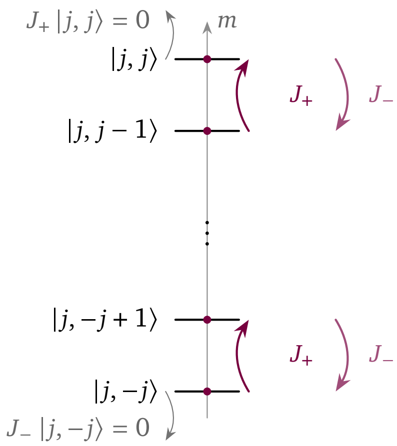

Figure 10.1: The weight ladder of the spin-j representation. The J_z-eigenvalues $m = j, j-1, \dots, -j$ label the rungs; the raising and lowering operators $J_\pm = J_x \pm \mathrm{i}J_y$ step between adjacent rungs, $J_\pm\,|j,m\rangle = \sqrt{j(j+1) - m(m\pm 1)}\;|j, m\pm 1\rangle$. The string is finite because J_+ annihilates the highest weight $|j,j\rangle$ and J_- the lowest $|j,-j\rangle$; matching the two terminations forces $2j \in \mathbb{Z}_{\geq 0}$ and fixes the dimension at $2j+1$. Half-integer j (e.g. $j = \frac{1}{2}, \frac{3}{2}$) are exactly the representations of $\mathfrak{su}(2)$ that fail to descend to SO(3).

Proof of everything except existence. Existence is constructive and gets its own section (10.3); here we show every irreducible has the stated form.

A top rung exists. Over $\mathbb{C}$ the operator J_z has an eigenvector, $J_z w = \lambda w$ — this is where algebraic closure is spent, and why Lemma 10.3 was bought. From (10.2),

$$J_z\,(J_+ w) = J_+ J_z\, w + J_+ w = (\lambda+1)\, J_+ w\ :$$

J_+ maps eigenvectors of J_z to eigenvectors one step higher (or to zero). Eigenvectors with distinct eigenvalues are linearly independent, and $\dim V < \infty$, so the chain $w,\ J_+w,\ J_+^2 w, \dots$ cannot stay nonzero forever: its last nonzero entry is a vector v_0 with

$$J_+\, v_0 = 0, \qquad J_z\, v_0 = j\, v_0$$

for some $j \in \mathbb{C}$ — a *highest-weight vector*. Nothing yet says j is real; that will be forced, not assumed.

Descend. Set $v_k = J_-^k v_0$. The lowering relation in (10.2) gives $J_z v_k = (j-k)v_k$. The raising coefficients follow by induction: $J_+ v_1 = J_+ J_- v_0 = (J_- J_+ + 2J_z)v_0 = 2j\, v_0$, and if $J_+ v_k = c_k v_{k-1}$ then

$$J_+\, v_{k+1} = (J_- J_+ + 2J_z)\, v_k = \big(c_k + 2(j-k)\big) v_k, \quad \text{so} \quad c_{k+1} = c_k + 2j - 2k.$$

With $c_1 = 2j$ this telescopes to $c_k = k(2j - k + 1)$, the coefficient in (10.3).

Terminate, and quantise. The nonzero v_k have distinct J_z-eigenvalues, hence are independent; so there is a smallest N with $v_{N+1} = 0$, $v_N \neq 0$. Apply J_+:

$$0 = J_+ v_{N+1} = (N+1)(2j - N)\, v_N \quad \Longrightarrow \quad j = \tfrac{N}{2}.$$

The highest weight is not merely real: it is a non-negative half-integer, extracted from pure algebra — no inner product, no Hermiticity, no physics was consulted.

Fill the space. The span of $v_0, \dots, v_N$ is invariant under J_z, J_+, J_- by (10.3), and these span $\mathfrak{sl}(2, \mathbb{C})$; being nonzero, by irreducibility the span is all of V. Hence $\dim V = N + 1 = 2j + 1$ and the J_z-spectrum is $\{j, j-1, \dots, -j\}$, each eigenvalue simple.

Uniqueness. The matrices in (10.3) depend only on j, so if V and V' are irreducible with the same j, the linear map $v_k \mapsto v'_k$ intertwines the actions of $J_z, J_\pm$, hence of all of $\mathfrak{sl}(2, \mathbb{C})$: an equivalence. Since $\dim V = 2j + 1$, dimension determines j. ■

Corollary 10.5 (The irreducible representations of SU(2)). *Differentiation followed by complexification (Lemma 10.3) is a bijection between representations of the group* SU(2) *and complex-linear representations of* $\mathfrak{sl}(2, \mathbb{C})$*, matching irreducibility, invariant subspaces and equivalence. In particular* SU(2) *has exactly one irreducible representation* Π_j *of each dimension* $2j + 1$*,* $j \in \frac{1}{2}\mathbb{Z}_{\geq 0}$*, and no others.*

Proof. Combine Lecture 9's dictionary for the connected, simply connected SU(2) with Lemma 10.3. ■

For the physics — and for any honest computation — the basis (10.3) is the wrong normalisation: the v_k are not unit vectors. Fixing this produces the most quoted formula of the subject.

Proposition 10.6 (Ladder normalisation). *Suppose V carries an inner product for which J_x, J_y, J_z are Hermitian — equivalently, for which the representation of* SU(2) *is unitary. (Such inner products exist: Section 10.3 exhibits one for every j.) Then $J_\pm^\dagger = J_\mp$, and the eigenvectors of J_z can be normalised and phased into an orthonormal basis $|j, m\rangle$, $m = j, j-1, \dots, -j$, with $J_z\,|j, m\rangle = m\,|j, m\rangle$ and*

$$J_\pm\,|j, m\rangle = \sqrt{j(j+1) - m(m \pm 1)}\;\; |j, m \pm 1\rangle\,. \tag{10.4}$$

Proof. $J_\pm^\dagger = (J_x \pm \mathrm{i}J_y)^\dagger = J_x \mp \mathrm{i}J_y = J_\mp$. Eigenvectors of the Hermitian J_z with distinct eigenvalues are orthogonal; normalise the highest one to $|j, j\rangle$ and define $|j, m-1\rangle = J_-\,|j, m\rangle \,/\, \|J_-\,|j, m\rangle\|$ downwards — this is the phase convention. Write $a_m = \|J_-\,|j, m\rangle\|^2$ and $b_m = \|J_+\,|j, m\rangle\|^2$. Then

$$a_m = \langle j, m|\, J_+ J_-\, |j, m\rangle = \langle j, m|\, (J_- J_+ + 2J_z)\, |j, m\rangle = b_m + 2m,$$

and applying J_+ to the definition of $|j, m-1\rangle$ gives $J_+\,|j, m-1\rangle = (J_- J_+ + 2J_z)\,|j, m\rangle \,/\, \sqrt{a_m} = \sqrt{a_m}\;|j, m\rangle$ (the middle step being $J_- J_+\,|j, m\rangle = b_m\,|j, m\rangle$: $J_- J_+$ commutes with J_z, so it preserves the one-dimensional weight space of $|j, m\rangle$ and acts there by a scalar, whose expectation value is $b_m = \|J_+\,|j, m\rangle\|^2$; together with $b_m + 2m = a_m$), i.e. $b_{m-1} = a_m$ — with a *positive* coefficient, so the same phases serve both ladders. The two relations give the downward recursion $a_m = a_{m+1} + 2m$ with seed $a_j = 2j$ (since $b_j = 0$), solved by $a_m = (j + m)(j - m + 1) = j(j+1) - m(m-1)$; then $b_m = a_{m+1} = j(j+1) - m(m+1)$. Taking positive square roots is the Condon–Shortley convention, and (10.4) is proved. ■

The formula every quantum mechanics course writes on day one of angular momentum is therefore a theorem about $\mathfrak{sl}(2,\mathbb{C})$, derived, with even the reality of the square roots guaranteed by the classification ($|m| \le j$).

Example 10.7 (The first rungs)**.** For $j = \frac{1}{2}$, formula (10.4) gives, in the basis $\left|\frac{1}{2}, \pm\frac{1}{2}\right\rangle$,

$$J_z = \tfrac{1}{2}\begin{pmatrix}1 & 0\\ 0 & -1\end{pmatrix}, \quad J_+ = \begin{pmatrix}0 & 1\\ 0 & 0\end{pmatrix}, \quad J_- = \begin{pmatrix}0 & 0\\ 1 & 0\end{pmatrix} \implies J_a = \tfrac{1}{2}\sigma_a :$$

the defining representation, recovered. For $j = 1$,

$$J_z = \begin{pmatrix}1 & 0 & 0\\ 0 & 0 & 0\\ 0 & 0 & -1\end{pmatrix}, \qquad J_+ = \sqrt{2}\begin{pmatrix}0 & 1 & 0\\ 0 & 0 & 1\\ 0 & 0 & 0\end{pmatrix}, \qquad J_- = J_+^{\mathsf{T}}.$$

Sheet 10 shows that this three-dimensional representation is equivalent to the adjoint representation of Lecture 9 — spin one is the algebra acting on itself — and builds the spin-$\frac{3}{2}$ matrices for practice.

10.3 A concrete model: homogeneous polynomials

Existence remains, and the construction that settles it is worth more than the settlement: it is the prototype of every "representation on functions" to come, including next lecture's spherical harmonics.

For $2j \in \mathbb{Z}_{\ge 0}$ let

$$V_j = \{\text{homogeneous polynomials of degree } 2j \text{ in } z_1, z_2\},$$

a complex vector space with basis $f_k = z_1^k z_2^{2j-k}$, $k = 0, \dots, 2j$, of dimension $2j+1$ — the right size already. The group acts by moving the variables, with Part I's inverse in place:

$$\big(\Pi_j(U)f\big)(z) = f\big(U^{-1}z\big), \qquad z = \left(\begin{smallmatrix}z_1\\ z_2\end{smallmatrix}\right), \quad U \in \mathrm{SU}(2). \tag{10.5}$$

This is a homomorphism (the two inverses compose in the right order, exactly as in Lecture 3), it preserves degree, and the matrix entries of $\Pi_j(U)$ are polynomial in the entries of $U^{-1} = U^\dagger$, hence continuous: a representation of SU(2) on V_j. Its derivative is a first-order differential operator, by the chain rule:

$$\big(\pi_j(X)f\big)(z) = \frac{\mathrm{d}}{\mathrm{d}t} f\big(e^{-tX}z\big)\Big|_{t=0} = -\sum_{a=1}^{2} (Xz)_a \frac{\partial f}{\partial z_a},$$

and the formula is complex-linear in X, so by Lemma 10.3 it also computes the extension to $\mathfrak{sl}(2,\mathbb{C})$. Feeding in H, E, F:

$$\pi_j(J_z) = \tfrac{1}{2}\big(z_2\partial_2 - z_1\partial_1\big), \qquad \pi_j(J_+) = -z_2\partial_1, \qquad \pi_j(J_-) = -z_1\partial_2,$$

so on the monomials

$$\pi_j(J_z)\, f_k = (j-k)\, f_k, \qquad \pi_j(J_+)\, f_k = -k\, f_{k-1}, \qquad \pi_j(J_-)\, f_k = -(2j-k)\, f_{k+1}.$$

The J_z-spectrum is $\{j, j-1, \dots, -j\}$, simple, and the monomial $f_0 = z_2^{2j}$ is a highest-weight vector of weight j.

Theorem 10.8 (The polynomial model is spin j). *Π_j is irreducible, and its derivative is equivalent to the spin-j representation of Theorem 10.4. In particular irreducible representations exist for every $j \in \frac{1}{2}\mathbb{Z}_{\geq 0}$, and the classification theorem is fully proved.*

Proof. Let $W \neq 0$ be an invariant subspace. It is invariant under $\pi_j(J_z)$, whose minimal polynomial has the $2j+1$ distinct roots $j-k$; the restriction to W divides it, so the restriction is diagonalisable and W contains an eigenvector of $\pi_j(J_z)$ — necessarily a multiple of some f_k, the spectrum being simple. The ladder coefficients above vanish only at the ends of the string, so repeated $\pi_j(J_\pm)$ carry f_k to nonzero multiples of every f_l: $W = V_j$. Irreducible with highest weight j, the derivative is the spin-j representation by the uniqueness clause of Theorem 10.4. ∎

The model also delivers, explicitly, the inner product that Proposition 10.6 hypothesised.

Proposition 10.9 (An invariant inner product, by Gaussian integral). *On polynomials in z_1, z_2 define*

$$\langle f, g\rangle = \frac{1}{\pi^2}\int_{\mathbb{C}^2} \overline{f(z)}\, g(z)\, e^{-|z_1|^2-|z_2|^2}\, \mathrm{d}V(z), \tag{10.6}$$

$\mathrm{d}V$ the Lebesgue measure on $\mathbb{C}^2 \cong \mathbb{R}^4$. This is an inner product; the monomials are orthogonal, with $\left\|z_1^k z_2^l\right\|^2 = k!\, l!$; and every $\Pi_j(U)$, $U \in \mathrm{SU}(2)$, is unitary for it.

Proof. Convergence is the Gaussian's gift. The integral factorises over the two coordinates, and in one variable, polar coordinates give

$$\frac{1}{\pi}\int_{\mathbb{C}} \bar{z}^k z^l\, e^{-|z|^2} \mathrm{d}A = \delta_{kl}\cdot 2\int_0^\infty r^{2k+1} e^{-r^2} \mathrm{d}r = \delta_{kl}\, k!$$

(the angular integral kills $k \neq l$; substitute $s = r^2$). Positivity and the monomial norms follow. For unitarity substitute $w = U^{-1}z$: a complex-linear map has real Jacobian $|\det_{\mathbb{C}}|^2$ (triangularise over $\mathbb{C}$; a 1×1 block $z \mapsto \lambda z$ has real Jacobian $|\lambda|^2$), here 1, and $|w_1|^2+|w_2|^2 = \|Uw\|^2 = \|w\|^2$, so the integral defining $\langle \Pi_j(U)f, \Pi_j(U)g\rangle$ is the integral defining $\langle f, g\rangle$. ∎

So unitarity is available for every spin — no group averaging was needed, only a Gaussian, an idea Lecture 13 will industrialise. (That *every* representation admits an invariant inner product is a different and stronger statement; it waits for Lecture 12.)

One accounting note before the Casimir. The polynomial model handed us group representations directly, so the existence half of Corollary 10.5 never actually drew on Lecture 9's lifting theorem: connectedness alone embeds the group list in the algebra list, and the model fills every slot. The blank cheque is cashed in earnest later — in Lecture 14, representations of the Lorentz algebra will arrive with no group action attached, and only simple connectivity will integrate them.

10.4 The Casimir operator and complete reducibility

Part I's sharpest small tool was Schur's lemma. Its algebra edition costs two lines.

Lemma 10.10 (Schur's lemma, algebra edition). *Let π be an irreducible complex-linear representation of a Lie algebra on V, and let $T \in \mathfrak{gl}(V)$ commute with every $\pi(X)$. Then $T = c\mathbb{1}$ for some $c \in \mathbb{C}$.*

Proof. T has an eigenvalue c, and $\ker(T - c\mathbb{1})$ is nonzero and invariant under every $\pi(X)$, hence equals V. (Lecture 4's proof of Schur's lemma; only the cast has changed.) ∎

Now a new character. Products such as J_x^2 are not elements of the Lie algebra — the abstract algebra knows only brackets — but any representation embeds the players in an associative algebra of operators, where products make sense. (The abstract home for such products, the universal enveloping algebra, is further reading; we will not need it.) Define, in any complex-linear representation of $\mathfrak{sl}(2,\mathbb{C})$, the *Casimir operator*

$$J^2 = J_x^2 + J_y^2 + J_z^2.$$

Proposition 10.11 (The Casimir). *In any complex-linear representation of* $\mathfrak{sl}(2,\mathbb{C})$*:*

(i)
$$J^2 = J_-J_+ + J_z(J_z+1) = J_+J_- + J_z(J_z-1); \tag{10.7}$$

(ii) J^2 *commutes with the image of the representation;*

(iii) on the irreducible spin-j representation, $J^2 = j(j+1)\mathbb{1}$.

Proof. (i) $J_\mp J_\pm = (J_x \mp \mathrm{i}J_y)(J_x \pm \mathrm{i}J_y) = J_x^2 + J_y^2 \pm \mathrm{i}\,[J_x, J_y] = J_x^2 + J_y^2 \mp J_z$, and add J_z^2. (ii) It suffices to commute with each J_a:

$$[J^2, J_a] = \sum_b \big(J_b[J_b, J_a] + [J_b, J_a]J_b\big) = \mathrm{i}\sum_{b,c} \epsilon_{bac}\,(J_bJ_c + J_cJ_b) = 0,$$

an antisymmetric ϵ contracted with a symmetric bracketless sum. (iii) By (ii) and Schur, $J^2 = c\mathbb{1}$; evaluate the first identity of (i) on the highest-weight vector: $c = 0 + j(j+1)$. ∎

The eigenvalue $j(j+1)$ — not j^2 — that every quantum course postulates for total angular momentum is therefore a two-line theorem. And Proposition 10.6 collapses in hindsight: $\|J_\pm\,|j,m\rangle\|^2 = \langle j,m|\,\big(J^2 - J_z(J_z \pm 1)\big)\,|j,m\rangle = j(j+1) - m(m\pm 1)$, one line, had the Casimir come first.

What about reducible representations? Part I's answer survives in two instalments, one payable now.

Theorem 10.12 (Complete reducibility). *(i) Every finite-dimensional unitary representation of any matrix Lie group is an orthogonal direct sum of irreducible subrepresentations.*

(ii) (Stated; proved in Lecture 12.) *Every representation of* SU(2) *on a finite-dimensional complex vector space admits an invariant inner product.*

Consequently every finite-dimensional representation of SU(2) *is equivalent to a direct sum of spin representations* Π_j.

Proof of (i). If V is not already irreducible, choose a proper nonzero invariant W; for $v \in W^\perp$ and $w \in W$, $\langle \Pi(g)v, w\rangle = \big\langle v, \Pi(g)^\dagger w\big\rangle = \big\langle v, \Pi(g^{-1})w\big\rangle = 0$, so $W^\perp$ is invariant too, and induction on dimension finishes — Lecture 3's argument, unchanged. ∎

Statement (ii) is Part I's averaging trick with the sum over the group replaced by an integral over SU(2); constructing that integral — the Haar measure — is Lecture 12's opening business, where Maschke's theorem returns in continuous dress. (A purely algebraic proof of complete reducibility also exists, with the Casimir doing the splitting; it is starred further reading.) Since every representation of $\mathfrak{su}(2)$ is the derivative of one of SU(2) (Corollary 10.5, resting on Lecture 9's lifting theorem and the simple connectivity of SU(2)), and Lemma 10.3 identifies complex-linear representations of $\mathfrak{sl}(2,\mathbb{C})$ with those of $\mathfrak{su}(2)$, the same decomposition holds for both.

10.5 The quantum theory of angular momentum and spin

A quantum system is rotationally symmetric when rotations act on its Hilbert space by unitaries commuting with the Hamiltonian — Lecture 9's quantum Noether setting. Acting by *which* group? The geometric answer is SO(3); the group this lecture has classified is SU(2). The discrepancy is real, productive, and deliberately deferred: quantum mechanics cares only about rays, and Lecture 11 will prove that ray-level representations of SO(3) are precisely ordinary representations of SU(2). For now we record the IOU and take the symmetry group to be SU(2), noting that at the level of observables there is nothing to choose: $\mathfrak{su}(2) \cong \mathfrak{so}(3)$ (Lectures 8 and 9), and the Hermitian generators obey $[J_a, J_b] = \mathrm{i}\,\epsilon_{abc} J_c$ either way.

Dividend one: multiplets. If $[H, \Pi(g)] = 0$, every energy eigenspace carries a representation of SU(2) (Lecture 9), which by Theorem 10.12 splits into spin-j irreducibles. Energy levels therefore come in multiplets of dimension $2j + 1 = 1, 2, 3, \ldots$; within a multiplet, J^2 and J_z commute with each other and with H, and the basis adapted to all three is the $|j, m\rangle$ of Proposition 10.6. The entire $|j, m\rangle$ bookkeeping of quantum mechanics — which states exist, what $J_\pm$ do to them, why degeneracies are $2j + 1$ — is Theorem 10.4 read aloud as physics.

Dividend two: spin. Lecture 9 proved that *orbital* angular momentum is integer-valued: the function action on wavefunctions $\psi(x)$ represents a U(1) along each axis, and the circle forces $m \in \mathbb{Z}$. The classification now exhibits what else is on offer: representations with half-integer j, unreachable by functions on space, perfectly available on an internal factor. A particle of spin s carries

$$\mathcal{H} = L^2(\mathbb{R}^3) \otimes \mathbb{C}^{2s+1},$$

rotations acting on the first factor by the function action and on the second by Π_s. This is the half-integers' home promised in Lecture 9. Why nature is *permitted* to occupy it — the address, so to speak — is still Lecture 11's.

Example 10.13 (Spin $\frac{1}{2}$, end to end)**.** The internal space is $\mathbb{C}^2$ with $J_a = \frac{1}{2}\sigma_a$; its states are *spinors*, spanned by $\left|\frac{1}{2}, \pm\frac{1}{2}\right\rangle$ ("spin up/down"). The operator rotating a spinor by angle θ about the unit axis $\hat{n}$ is

$$U(\theta, \hat{n}) = e^{-\mathrm{i}\theta\,\hat{n}\cdot\vec{J}} = e^{-\mathrm{i}\frac{\theta}{2}\,\hat{n}\cdot\vec{\sigma}} = \cos\tfrac{\theta}{2}\,\mathbb{1} - \mathrm{i}\sin\tfrac{\theta}{2}\;\hat{n}\cdot\vec{\sigma},$$

Lecture 8's exponential formula, re-employed as the daily tool of magnetic resonance. The halved angle, decorative in Lecture 8 and structural in Lecture 9, is about to become physics.

Example 10.14 (Larmor precession: the group action in real time)**.** A magnetic moment $\vec{\mu} = \gamma\vec{J}$ in a uniform field $\vec{B} = B\hat{z}$ has Hamiltonian $H = -\vec{\mu}\cdot\vec{B} = \omega J_z$ with $\omega = -\gamma B$. The evolution operator

$$U(t) = e^{-\mathrm{i}\omega t\,J_z} = U(\omega t, \hat{z})$$

is the rotation operator about the field, arriving in real time: the dynamics *is* a one-parameter subgroup of the symmetry group. In the Heisenberg picture the spin operators precess,

$$J_x(t) = \cos(\omega t)\,J_x - \sin(\omega t)\,J_y, \qquad J_y(t) = \sin(\omega t)\,J_x + \cos(\omega t)\,J_y,$$

(differentiate and use the brackets), so $\langle\vec{J}\,\rangle$ circles the field at the Larmor frequency $|\omega|$. The conjugation $A \mapsto U^\dagger A U$ driving this is Lecture 9's adjoint representation, run by the clock — Lecture 9's adjoint-representation formula earning a salary.

Example 10.15 (Stern–Gerlach: counting a dimension)**.** Send a beam of atoms through a strongly inhomogeneous field. The force on a moment is proportional to m, the J_z eigenvalue, so the beam splits into one spot per eigenvalue: $2j+1$ spots. Silver atoms give *two*. Two is $2j+1$ for $j=\frac{1}{2}$ and for no integer ℓ — so the splitting cannot come from orbital motion, which Lecture 9 confined to integers. Read with this lecture's eyes, the 1922 experiment counts the dimension of an irreducible representation, finds it even, and thereby discovers the internal factor: spin $\frac{1}{2}$ was forced, not fitted.

One computation remains: the cliffhanger.

Proposition 10.16 (A full turn is not the identity)**.** *For every j,*

$$\Pi_j(-\mathbb{1}) = (-1)^{2j}\,\mathbb{1}. \tag{10.8}$$

Proof. By Lecture 9, $-\mathbb{1} = e^{2\pi X_3}$ is the endpoint at $\theta = 2\pi$ of the path $e^{-\mathrm{i}\theta\sigma_3/2}$, whose image under Ad is the full turn $R_z(2\pi) = \mathbb{1}$. Then $\Pi_j(-\mathbb{1}) = e^{2\pi\,\pi_j(X_3)} = e^{-2\pi\mathrm{i}J_z}$ has eigenvalues $e^{-2\pi\mathrm{i}m}$; every m differs from j by an integer, so each eigenvalue equals $(-1)^{2j}$ and the operator is that scalar. ∎

For integer j the full turn acts trivially, and all is calm. For half-integer j it acts as $-\mathbb{1}$: rotating a spinor by 2π flips its sign, and only a 4π turn restores it. Two reactions are available. The alarmed one — a 2π rotation does nothing to space, so an operator that does something cannot represent rotations — is correct, and its resolution is that $\Pi_{1/2}$ represents not SO(3) but SU(2), where the path of exponentials closes only after 4π. The relaxed one — a global sign is invisible on a single ray, so quantum mechanics may not care — is also correct, but incomplete: put the rotated spinor in superposition with an unrotated companion and the sign interferes observably. (Neutron interferometers have measured exactly this; Lecture 11 reports the experiment.) Whether the relaxed reaction can be promoted to a principle — which "representations up to phase" quantum mechanics really permits, and why SU(2) is the lawful answer to SO(3)'s question — is precisely Lecture 11's business.

So the season's first harvest is in: a complete classification, a concrete model, an honest Casimir, and a physics — multiplets, ladders, precession, spin — that turned out to be representation theory wearing a lab coat. Lecture 11 takes the same algebra to the group SO(3): which Π_j descend through the kernel $\{\pm\mathbb{1}\}$, where the integer representations live as functions on the sphere, and how rays rescue the rest. Lecture 12 then lets angular momenta meet: tensor products, Clebsch–Gordan, and the return of characters.

Remark 10.17 (Computer algebra)**.** Notebook 10 generates the $(2j+1)$-dimensional matrices J_z, $J_\pm$ from (10.4) for arbitrary j; verifies the brackets (10.2) and $J^2 = j(j+1)\,\mathbb{1}$ to machine precision; exponentiates with `MatrixExp` to rotation operators, confirming unitarity and the sign (10.8); and animates a precessing Bloch vector against the spinor's half-frequency phase, making the 4π period visible.

Summary.

- *Classification.* For each $j \in \frac{1}{2}\mathbb{Z}_{\geq 0}$ there is, up to equivalence, exactly one irreducible representation π_j of $\mathfrak{sl}(2,\mathbb{C})$, of dimension $2j+1$, with J_z-spectrum $\{-j, -j+1, \ldots, j\}$ (Theorem 10.4, eq. (10.3)).
- *Ladder operators.* $J_\pm\,|j,m\rangle = \sqrt{j(j+1) - m(m\pm 1)}\;|j, m\pm 1\rangle$, with J_+ and J_- annihilating the highest and lowest weights (eq. (10.4)).

- *Polynomial model.* π_j is realised on the homogeneous polynomials of degree $2j$ in two variables, unitary for an explicit SU(2)-invariant inner product.
- *Casimir.* $J^2 = J_x^2 + J_y^2 + J_z^2$ acts as the scalar $j(j+1)\mathbb{1}$ on π_j.
- *Central element.* $\Pi_j(-\mathbb{1}) = (-1)^{2j}\mathbb{1}$, so the half-integer representations do not descend to SO(3) (eq. (10.8)).

Tutorial. Exercise Sheet 10.

References

Hall §4.2, §4.6 (sl$(2,\mathbb{C})$ representations); Fulton–Harris §11.1; Woit (the chapters on SU(2) and spin); Tung (SO(3)/SU(2)); Jones Pt. 8–9.

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